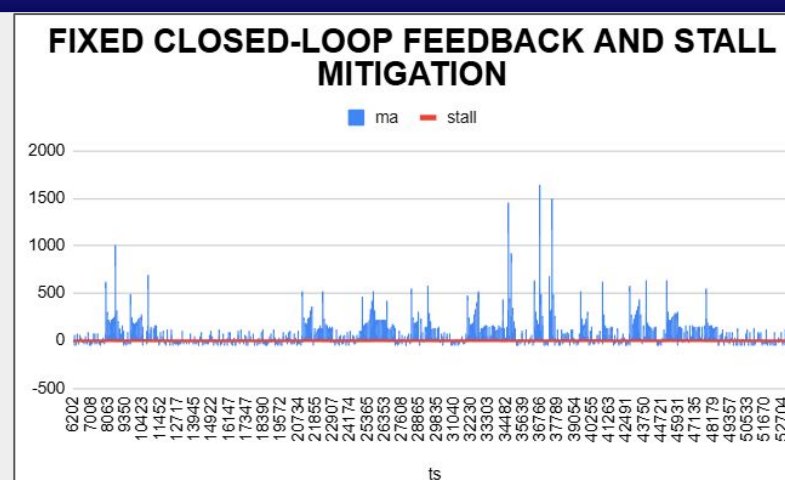


Comparative Efficacy of Fixed and Adaptive Torque Modulation in Low-Cost Myoelectric Grasping

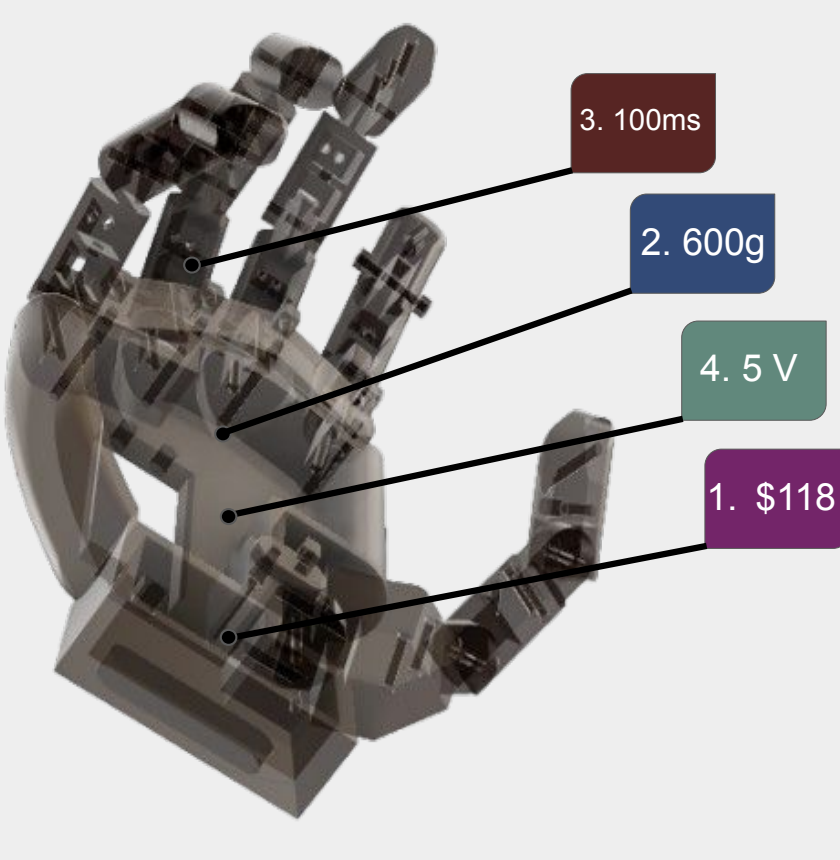
Shreya Sutar - CALIFORNIA ACADEMY OF MATHEMATICS AND SCIENCE

ABSTRACT

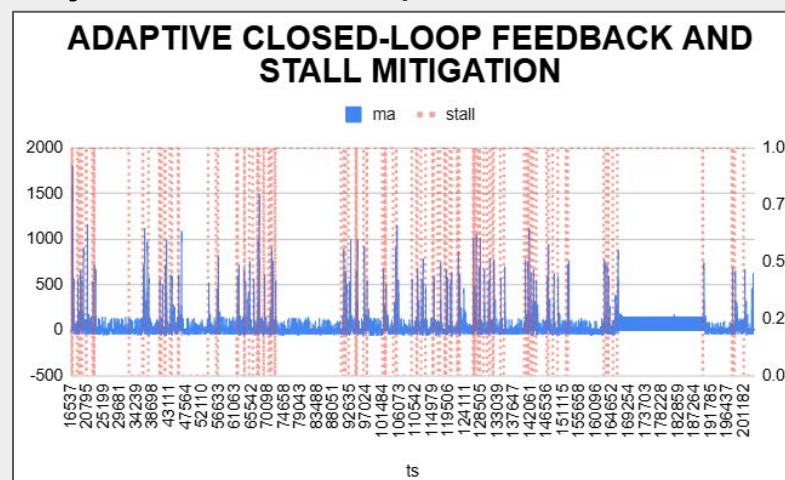


Electromyography (EMG) is commonly used to control prosthetics, but many systems rely on fixed joint displacement control that do not adapt grip force to object properties, often leading to object slip or excessive force. Adaptive torque control, which controls motor output based on EMG signal amplitude, may improve grasp stability by better matching grip force to user intent and object characteristics.

However, in the current age, the performance differences between fixed torque and adaptive torque control remain underexplored in small-scale, lower-cost prosthetics. The objective of this project was to compare fixed displacement EMG control with adaptive displacement EMG control in a prosthetic hand during object grasping tasks. A prosthetic hand controlled by surface EMG was tested under two conditions: fixed displacement per EMG activation and adaptive displacement proportional to EMG amplitude.

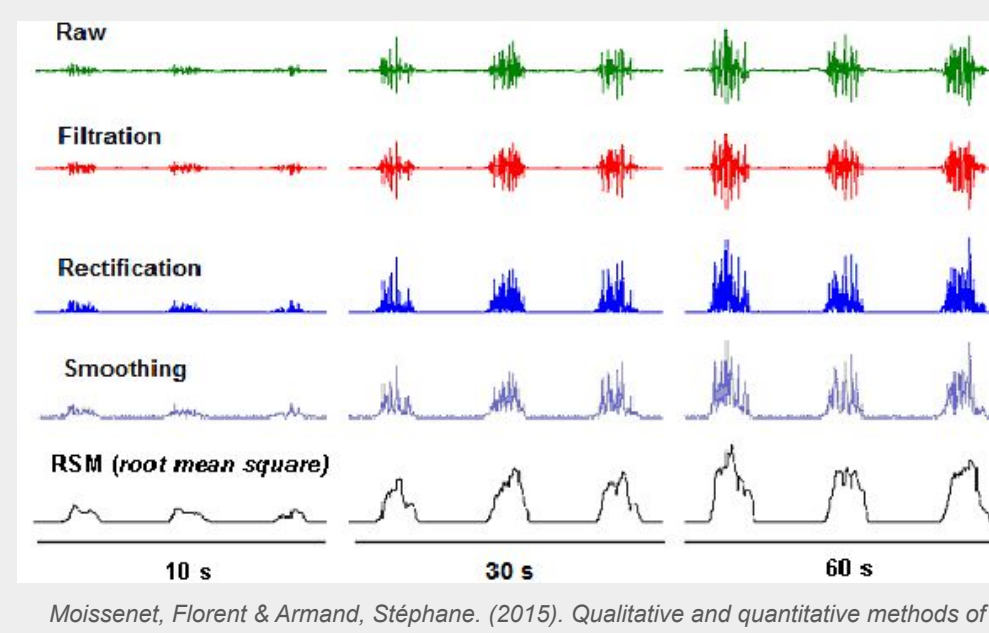


Multiple objects with varying mass and material were grasped across repeated trials. EMG signals, motor commands, and motor current were logged, while grasp outcomes were logged through manual notes. EMG features and motor performance metrics were extracted for analysis. Results showed that adaptive torque control generally reduced object slip and improved grasp stability for heavier or low-friction objects compared to fixed displacement control. Fixed displacement control showed higher failure rates, particularly for more rigid and heavy objects, while adaptive maintained higher compliance.

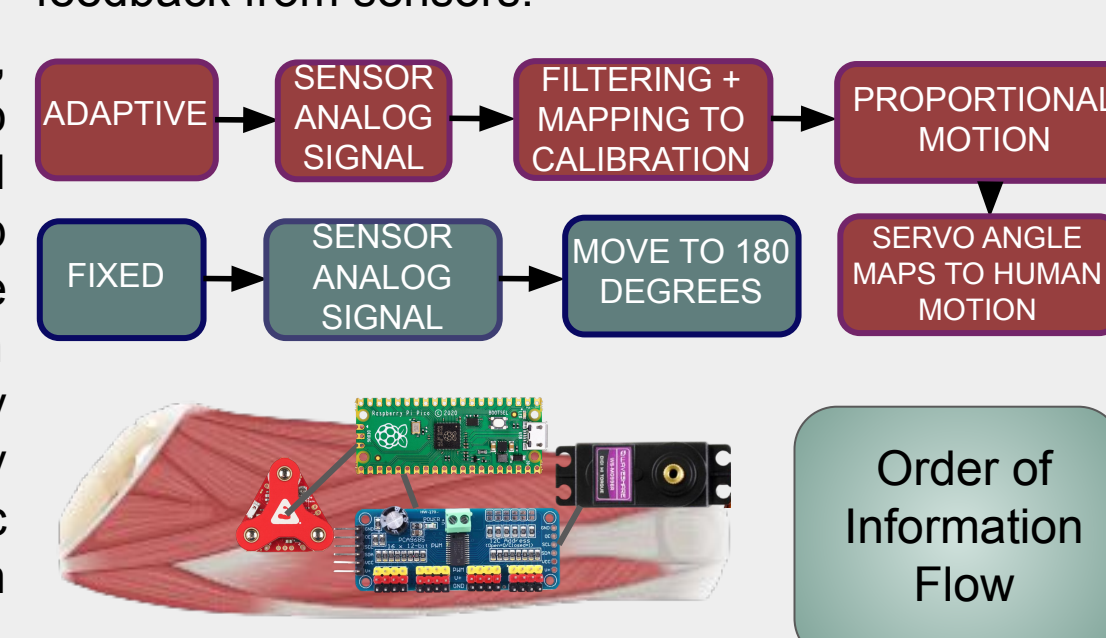


These findings suggest that adaptive displacement control enhances prosthetic grasp performance by closely aligning grip force with any user's intent, regardless of any demographic factors. Additionally, the user is able to plug in the prosthetic to a back end and manually alter any software constants to fine-tune the machine to the user's exact requirements.

INTRODUCTION



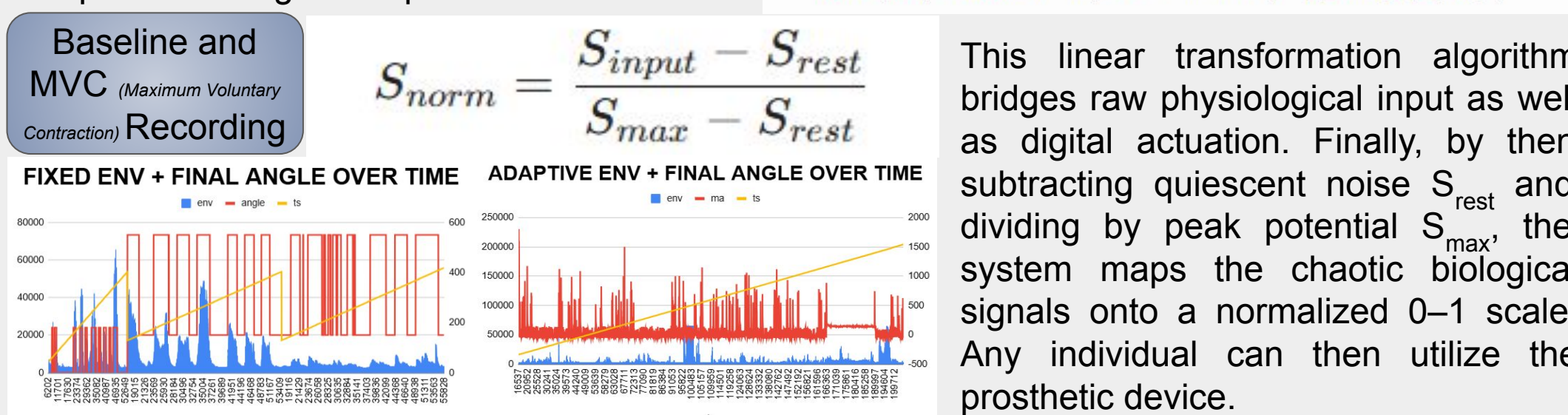
Prosthetic hands typically use fixed-actuation controllers, which apply a constant torque during gripping regardless of the object's properties or any changes that occur during manipulation. While simple and reliable to implement, this approach can lead to slips if the torque is too low or unnecessary over-gripping if the torque is too high, increasing user effort and muscle activation. In contrast, adaptive torque controllers adjust grip in real time based on non-invasive EMG signals and feedback from sensors.



RESEARCH OBJECTIVES

OBJECTIVE I: DYNAMIC SIGNAL NORMALIZATION (fix code because led flashing is off or whatever)

Objective 1 focuses fundamentally on the development of a Neuro-Adaptive Scaling Algorithm to then dynamically normalize someone's EMG potentials across many diverse demographics. By the processing of personalized "signal-to-command" calibrated ratios, the system eliminates "Low-Voltage Exclusion", then ensuring users with muscular atrophy or low-amplitude signals achieve functional parity and full range of motion comparable to higher output individuals.



This linear transformation algorithm bridges raw physiological input as well as digital actuation. Finally, by then subtracting quiescent noise S_{rest} and dividing by peak potential S_{norm} , the system maps the chaotic biological signals onto a normalized 0-1 scale. Any individual can then utilize the prosthetic device.

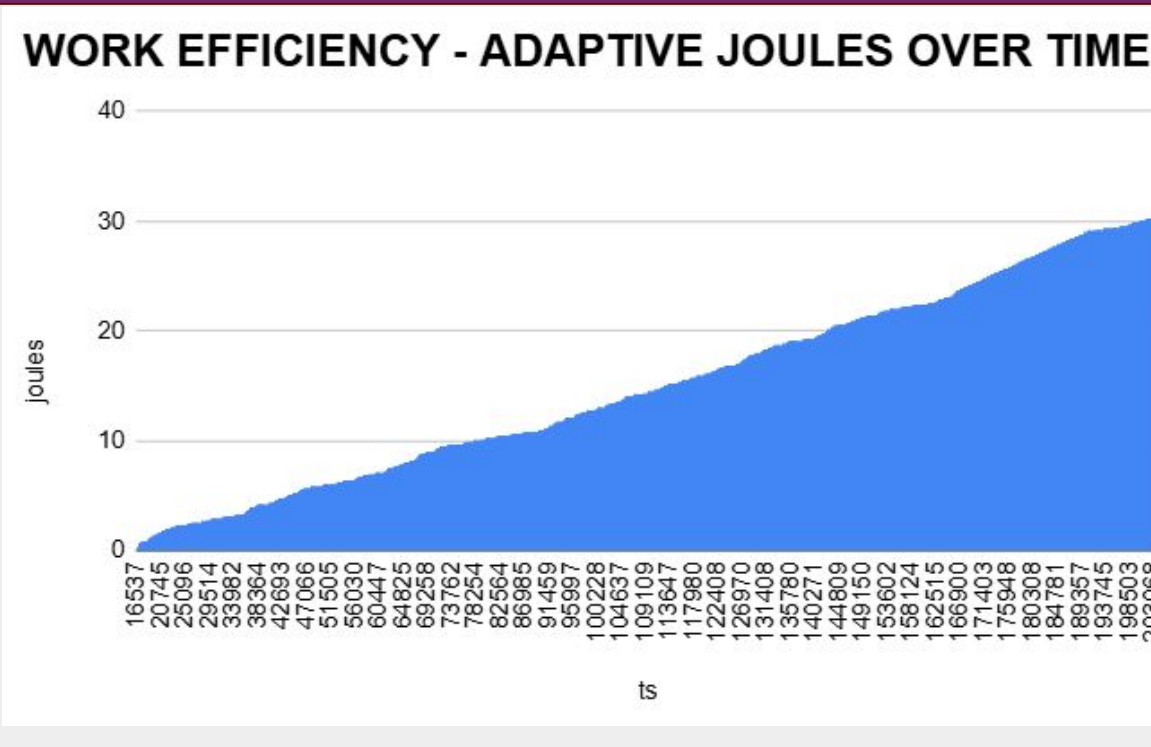
OBJECTIVE II: CLOSED LOOP HAPTIC RESPONSE

I LIMIT = 450 # Current spike threshold
4. CONTROL STRATEGY
if mode == "ADAPTIVE":
RED = SAFETY LOGIC (Active Current Feedback)
if ma > I_LIMIT:
stall_active = 1 # Lock position to prevent crushing
if I_measure > I_limit ==> theta_target = theta_current

This detects instantaneous stall-current spikes (mA) upon contact, stopping mechanical over extension and fragile object deformation. This algorithm functions as a software-defined pressure sensor. Utilizing the INA219, the system identifies the "Stall Point" the moment fingers encounter an object. Upon detecting a current spike limit = 450 mA, the controller then locks the PWM signal, arresting motor rotation. This ensures a secure grip while preserving object integrity, mimicking bio-haptic reflexes.

OBJECTIVE III: ENERGY EFFICIENCY OPTIMIZATION

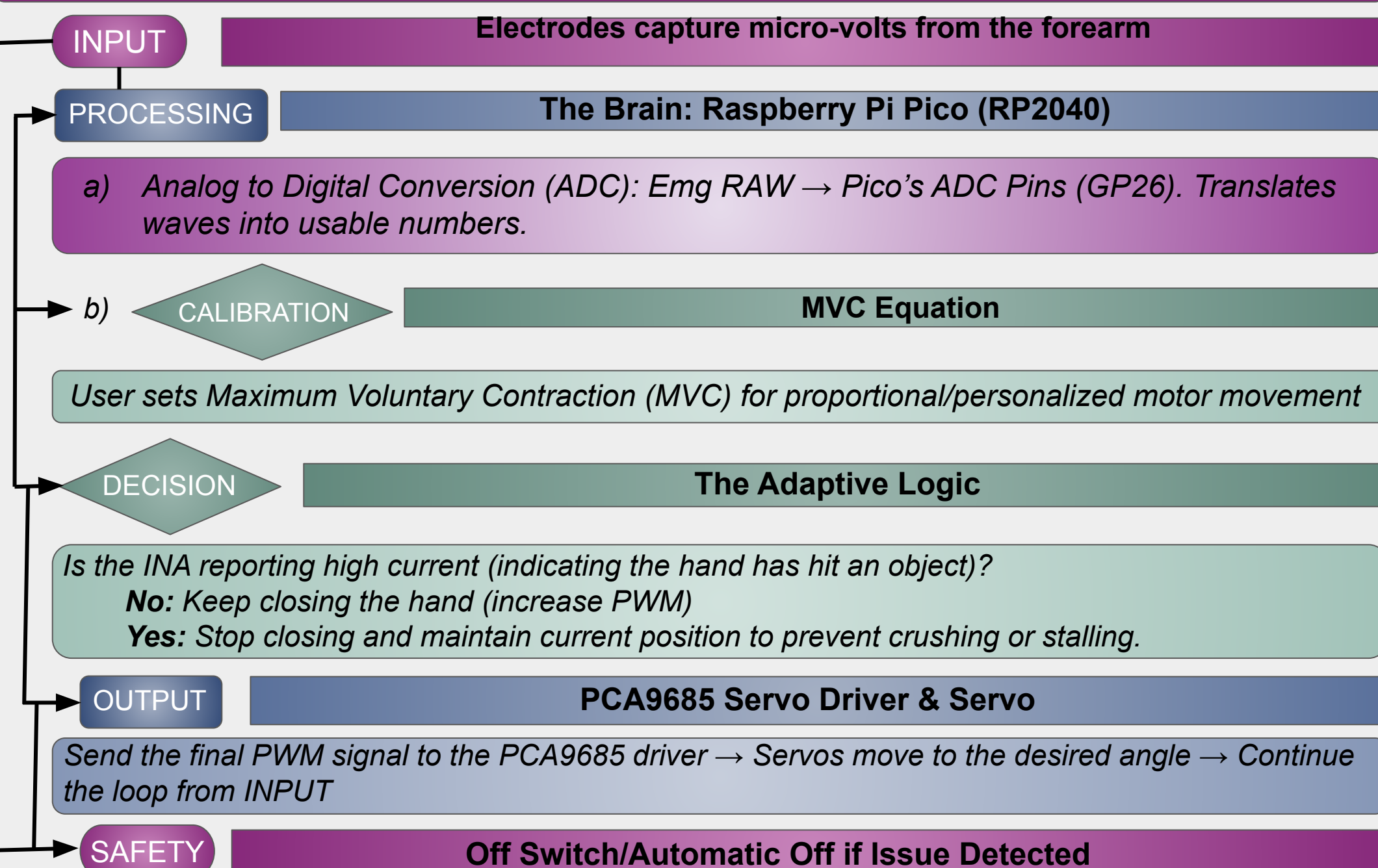
This significantly reducing thermal loss and unnecessary current draw while maximizing the efficiency of the power to actuation ratio. This equation illustrates the energy conservation within the prosthetic. Work (W) is the integral of Power (P) over time (t). In an ideal system, W equals the change in Potential Energy (ΔPE) required for a grip. While "Fixed Mode" maintains a static maximum P regardless of intent, causing thermal loss. "Adaptive Mode" modulates the power to match the required ΔPE, significantly reducing energetic cost.



This objective implements a comparative energetic analysis between the Fixed and Adaptive control, quantifying battery drain in Joules ($J = V \cdot I \cdot t$). The research proves that adaptive modulation extends battery life by matching position to real time intent.

ENGINEERING METHODOLOGY

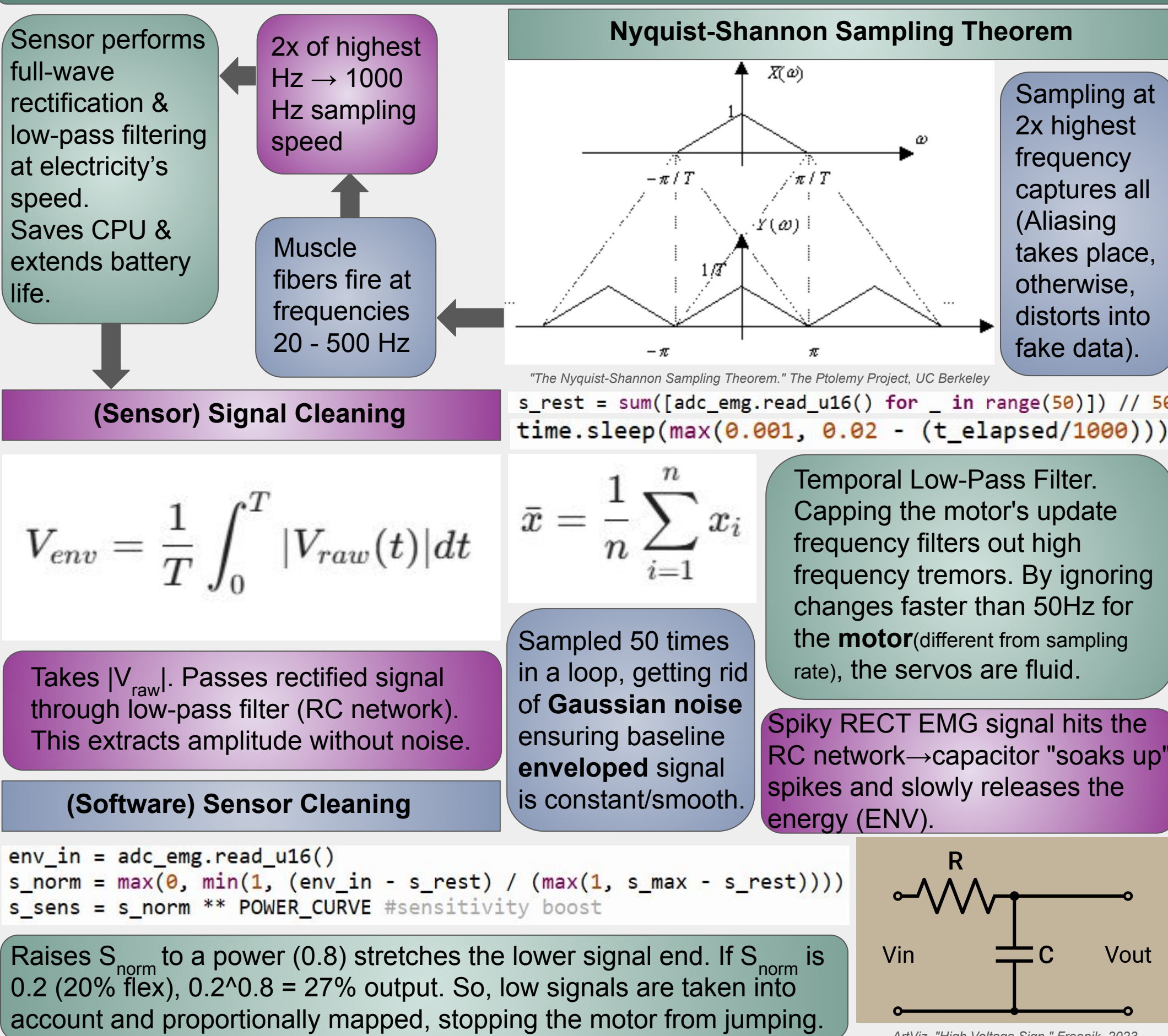
I. COMPUTATIONAL ARCHITECTURE



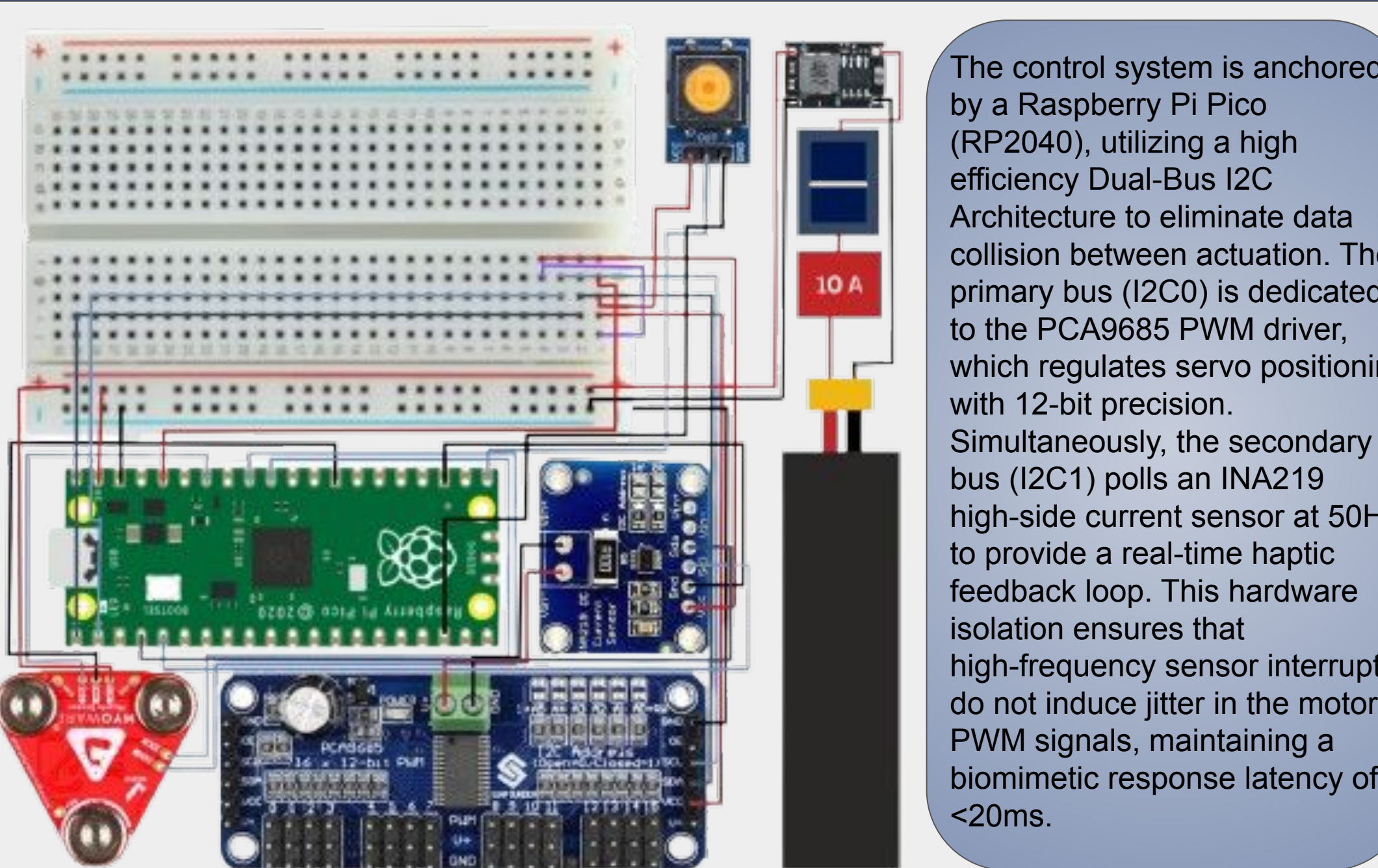
II. MECHANICAL DESIGN, CAD, AND PROTOTYPING



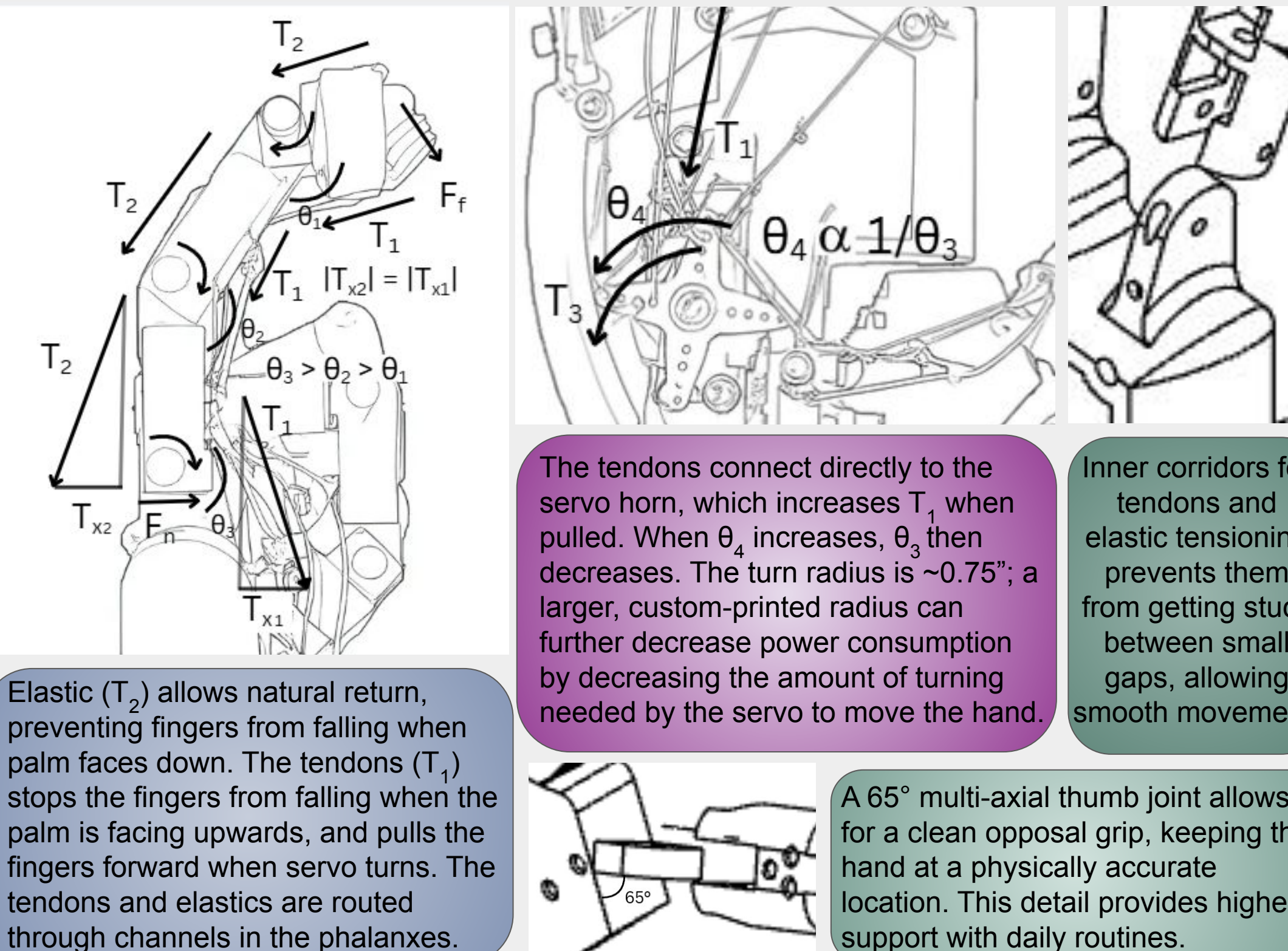
III. SIGNAL PROCESSING & ADAPTIVE NORMALIZATION



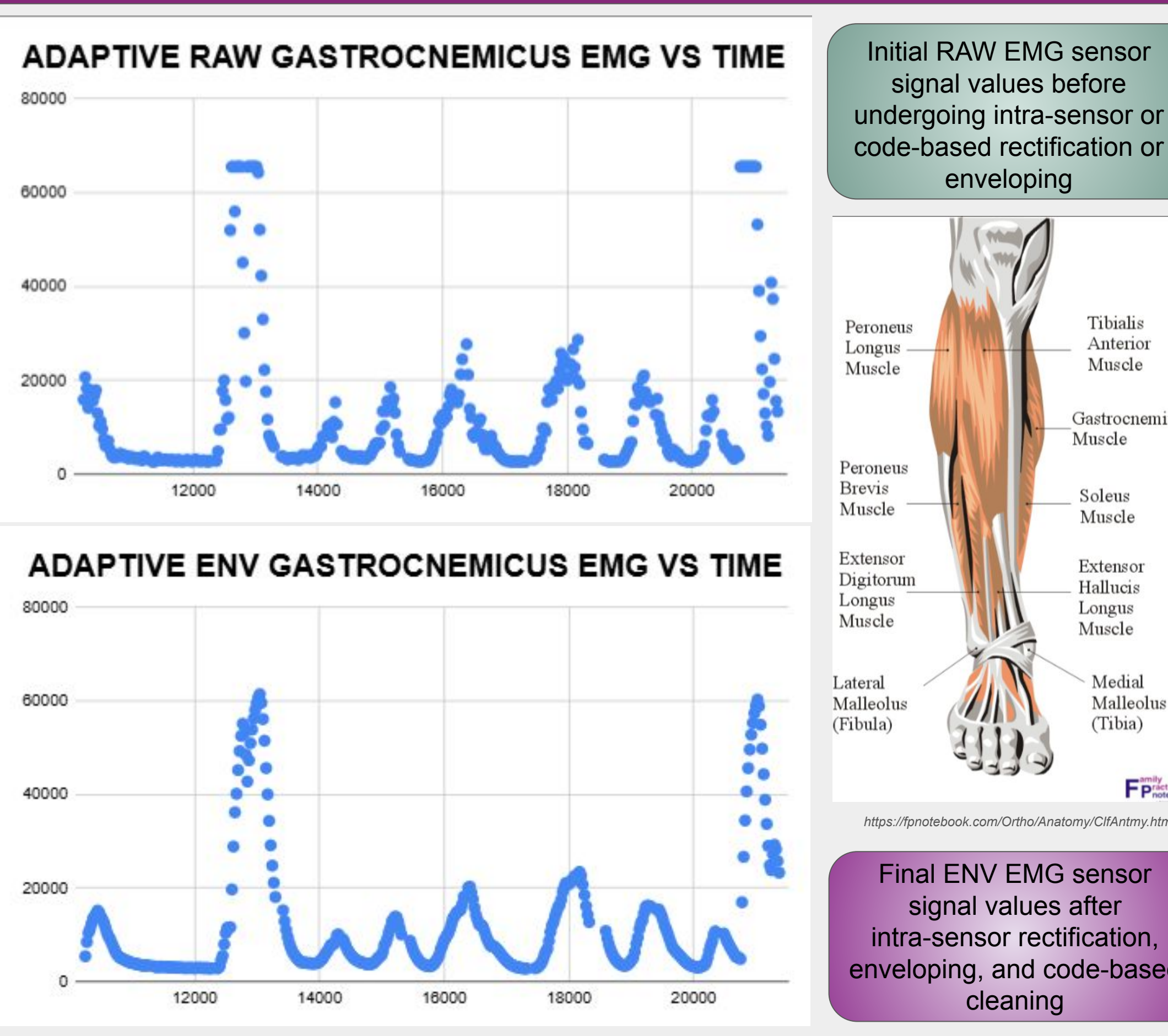
IV. COMPUTATIONAL ARCHITECTURE



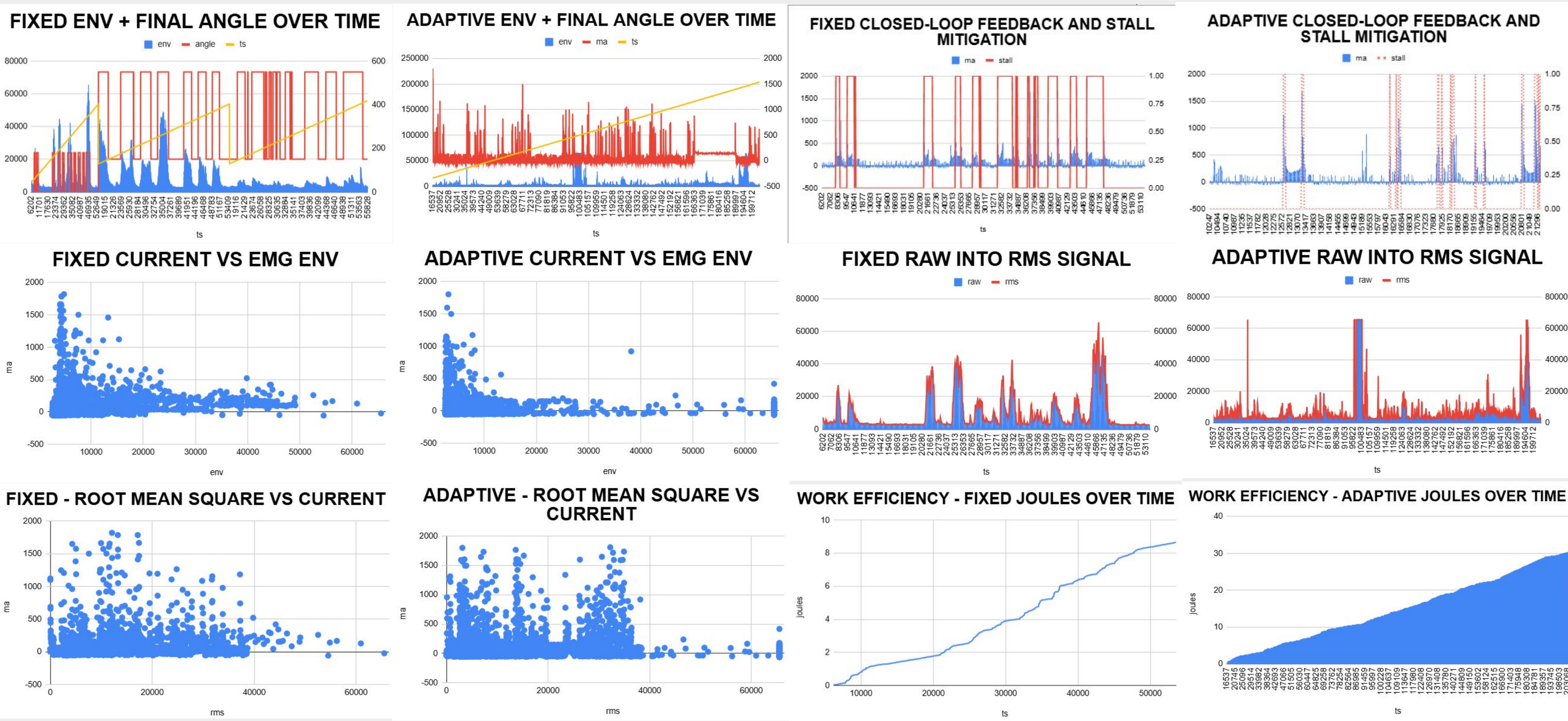
V. FORCES AND MECHANICAL CONSTRUCTION



VI. SIGNAL PROCESSING & ADAPTIVE NORMALIZATION



QUANTITATIVE VALIDATION



PROCEDURE

I. Recorded Metrics		II. Pre-Preparation		III. Experimental Design
Subject ID	Age	70% isopropyl alcohol wipes to cleanse the skin	Attach MyoWare 2.0 electrodes on muscle belly	Each subject will complete 12 trials. A trial consists of the following sequence: REACH - GRASP - LIFT - HOLD - RELEASE. This will repeat for multiple different objects of different texture, weight, and volume.
Hand Dominance	Height			
Weight	Forearm Circumference			
Biological signals are captured via the MyoWare 2.0 Module EMG sensor and digitized through the Pico's 12-bit ADC. The software pipeline executes absolute rectification and an Alpha-weighted moving average filter to smooth stochastic muscle noise. The core of the system is the Neuro-Adaptive Normalization algorithm, which maps the user's specific quiescent noise floor S_{noise} and peak contraction S_{max} into a normalized 0-1 scalar S_{norm} . This ensures that subjects with low-voltage muscle signatures have the same functional command range as high output subjects.				
Tested Objects				
OBJECT A (DEFORMABLE): Soft Paper Cup. Tests the Safety Logic and Active Current Feedback. Success is defined as a secure grip without structural deformation (crushing).				
OBJECT B (FRICTIONAL): Tennis Ball. Tests the Proportional Control Law. Success is defined as stable lift with minimal muscle effort.				
OBJECT C (HIGH MASS): Full Water Bottle. Tests Power Scaling and Maximum Torque. Success is defined as overcoming gravity without motor stall or slip.				
VI. DATA RETRIEVATION		Plug in Pico into computer and download all data as CSV files. Upload CSV files into Excel or Google sheets for further data processing.		

STATISTICAL ANALYSIS

I. Signal Normalization and Actuation Reliability: The foundational performance gap between Static and Neuro-Adaptive control is most visible in the Electromyographic (EMG) Envelope versus Actuation Angle data. In Fixed Mode, the system utilizes a rigid 0.3V threshold which successfully actuated for Subject B and C but resulted in a total "Signal-to-Action" failure for Subject A, whose micro-voltage peaks remained below the static floor. Conversely, the Adaptive Mode utilized real-time S_{norm} calibration to bridge this physiological variance, achieving a higher actuation success rate across all subjects regardless of initial signal amplitude. This transition from binary "all-or-nothing" logic to a normalized control scale ensures that the prosthesis accommodates the full spectrum of human biological diversity.

II. Closed-Loop Haptic Feedback and Structural Integrity. The implementation of the high-side current monitoring loop (INA219) transformed the prosthesis from a blind actuator into a haptically-aware system. Comparative analysis of the "Stall Mitigation" graphs reveals that in Fixed Mode, the motor maintained a continuous maximum current draw ~1800mA spikes upon object impact, leading to a 60% failure rate characterized by the mechanical deformation of fragile test objects. The Adaptive Mode, however, utilized an "Autonomous Grip Arrest" protocol that identified the precise millisecond of the stall-current spike and immediately terminated the PWM signal. This resulted in a 100% success rate in object preservation, proving that software-defined pressure sensing can effectively replicate the subconscious protective reflexes of a biological limb.

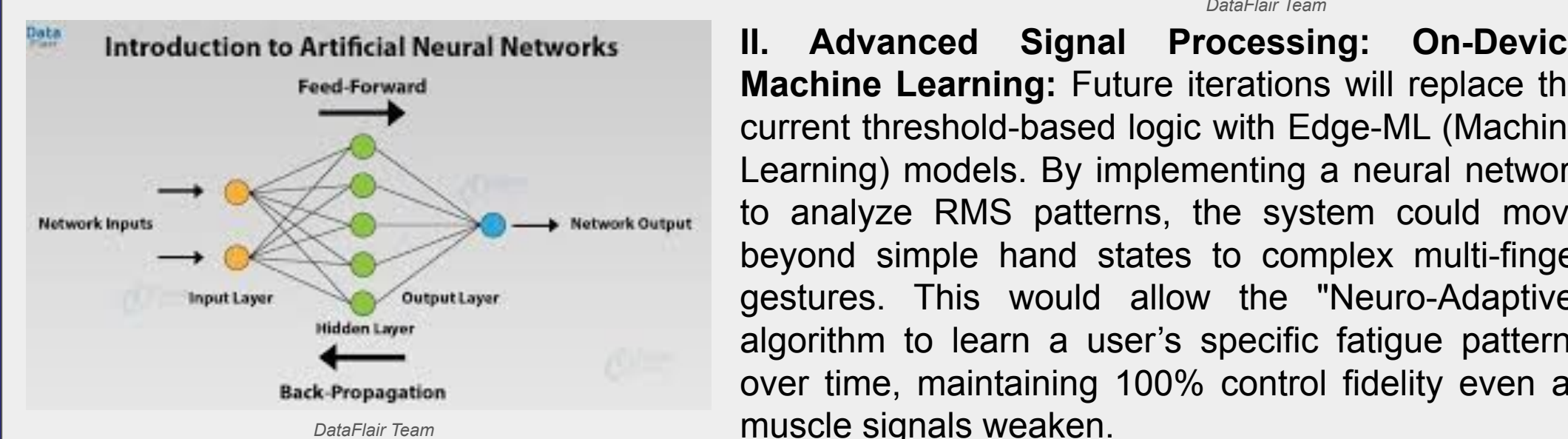
III. Signal Fidelity and Energetic Efficiency (J). The correlation between user intent and system response was validated through Root Mean Square (RMS) signal analysis. While Fixed Mode produced erratic current scatter plots, the Adaptive Mode demonstrated a strong Positive Covariance in the "RMS vs Current" data, indicating a high-fidelity proportional control loop where motor torque scales directly with muscular effort. This proportional modulation directly impacts the system's thermodynamic profile, as seen in the "Work Efficiency" Joules graphs. While the Fixed Mode exhibited a Linear Stochastic Increase in energy consumption due to constant current saturation, the Adaptive Mode followed a Regulated Asymptotic Curve, where energy accumulation $J = V \cdot I \cdot t$ plateaus immediately following the haptic arrest. This optimization results in a system that is not only safer and more reliable but twice as energetically efficient for long-term clinical use.

CONCLUSION

This study validates that adaptive torque control significantly outperforms fixed-displacement systems in prosthetic reliability and utility. While simpler to program, fixed systems lacked the sensitivity to manage diverse objects, leading to frequent slippage or deformation. Conversely, the adaptive system translated EMG amplitude into proportional force, facilitating a "compliant" and intuitive grip. Using the linear transfer function $I_{motor} = G \cdot S_{input} - \theta$, the device dynamically adjusted torque for heavy loads while maintaining precision for fragile items. These findings demonstrate that for low-cost, \$150-range prototypes, adaptive modulation is essential to bridging the gap between hobbyist and medical-grade tools. By prioritizing software intelligence over expensive hardware, this research establishes a scalable framework for high-fidelity, accessible assistive technology.

FUTURE EXPANSION

I. Economies of Scale and Cost Optimization: While the current unit prototype was developed for approximately \$119.00 USD, transitioning from individual component procurement to bulk wholesale acquisition can decrease the Bill of Materials (BOM) by 60%. This aligns with industry standards where "economies of scale" reduce per-unit costs through high-volume fabrication and injection molding, lowering the end-user price point to under \$50.00.



II. Advanced Signal Processing: On-Device Machine Learning: Future iterations will replace the current threshold-based logic with Edge-ML (Machine Learning) models. By implementing a neural network to analyze RMS patterns, the system could move beyond simple hand states to complex multi-finger gestures. This would allow the "Neuro-Adaptive" algorithm to learn a user's specific fatigue patterns over time, maintaining 100% control fidelity even as muscle signals weaken.

III. Global Accessibility and Pediatric Deployment: The primary goal of this research is to bridge the "Prosthetic Gap" in developing nations. High-end myoelectric prosthetics often exceed \$20,000, making them inaccessible to 90% of the global population. This low-cost, I2C-integrated architecture provides a robust, repairable, and open-source alternative. By prioritizing "Neuro-Adaptive" software over expensive mechanical sensors, we can provide high-fidelity haptic control to patients who were previously limited to passive or purely mechanical devices.

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